MIL-T-14981A(MU)
3 April 1970

SUPERSEDING
MIL-T-14981(MU)
23 September 1968

MILITARY SPECIFICATION

TOOL, SHELL REMOVAL ASSEMBLIES: H4172 AND XH4196

1. SCOPE

1.1 This specification covers special requirements, quality assurance provisions, and the preparation for delivery criteria for one type of shell removal tool used to remove XM454 and M422 projectiles from the artillery weapon.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning and Preparation of Ferrous and Zinc Coated Surfaces for Organic Protective Coating.

MILITARY

MIL-B-2427 - Boxes, Ammunition Packing, Wood, Nailed.
MIL-I-6866 - Inspection Penetrant, Method of.
MIL-I-6868 - Inspection Process, Magnetic Particle.
MIL-S-69902(MU) - Soldering Electrical Connections for Nuclear Weapons Systems, Procedure for.

STANDARDS

FEDERAL

Federal Test Method Standard No. 151
MIL-STD-1261 - Welding Procedures for Structural Steel

FSC-1190

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.
MS-35771 - Bearing, Sleeve.

DRAWINGS

ORDNANCE CORPS

8797272 - Marking Diagram and Sealing for Rectangular Containers
8839639 - Wrench, Torque
8839640 - Plug and Rod Assembly
8839641 - Plug
8839642 - Rod
8857225 - Tool, Shell Removal Assembly: H4172
8857302 - Pin
8857427 - Plug and Rod Assembly
8857430 - Plate, Support
8857514 - Plug
8857515 - Rod
8862510 - Bearing, Roller, Thrust
8862511 - Support Plate Assembly
8862512 - Retainer
8862525 - Threaded Transducer Assembly
8862526 - Transducer, Threaded
8862527 - Gusset
8862529 - Wrench, Bearing
8863247 - Box, Pkg, Ammo, for Tool, Shell Removal Assy: H4172
8863848 - Box, Pkg, Ammo, for Tool, Shell Removal Assy: XH4196
8863343 - Rod, Extension
8863402 - Retainer Assembly
8864331 - Tool, Shell Romoval Assy: XH4196
ET-8857225 - Equipment Tabulation (H-4172)
ET-8864331 - Equipment Tabulation (XH4196)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material used to fabricate any part complies with the specifications listed on the applicable drawing.

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3.2 Assembly. The shell removal tool shall comply with all requirements specified on Drawing (dwgs) 8857225 or 8864331 as applicable with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Static load capacity of bearing (see dwg 8862510). The roller bearing shall not have a permanent deformation of any roller at point of contact more than 0.0001 inches of the roller diameter when subjected to a static load of 120,000 pounds minimum, when tested in accordance with 4.4.2.

3.5 Functional test (see dwgs. 88632A7 and 88632A8). The shell removal tool shall fit snugly in the box with a minimum of looseness or play when tested in accordance with paragraph 4.4.3.

* 3.6 Paint adhesion (see dwgs 8862512, 8857430, 8857427, 8862525, 8862529, 8839639, 88632A7, 8839640, 88638A8). Paint adhesion shall comply with Paint Adhesion, section 3, TT-C-490.

3.7 Workmanship.

3.7.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.7.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.7.3 Burrs. Parts shall be free from burrs.

3.7.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.7.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted (see 6.5). When the contract requires first article inspection, the contractor shall submit one complete assembly (see dwgs 8857225 or 8864331 as applicable), and one set of parts, for first article testing to the testing facility designated in the invitation for bids or request for proposal.

* 4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests need not be conducted in order specified in Table I.

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Table I

First article inspections and tests

Examination or test	Contractor's inspection before submission	Non-Destructive	Parts	Sub-Ass'y	Ass'y
1. Classification of characteristics (see 4.3.2.1)	X	X	X	X	X
2. Static load capacity of bearing (see 3.4)	X	X	X		
3. Functional test (see 3.5)	X	X			X
4. Tensile (see 4.3.3.1)	X	X	X		
5. Welding (see 4.3.3.3)	X	X		X	
6. Paint adhesion (see 3.6)	X	X	X	X	X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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* Table II

Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	1121- ∞	45
26-27	22	48-51	30	113-131	38		

* Table III

Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	274- ∞	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Retainer (see dwg 8862512 covering a detail of dwg 8863402) (before painting).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Width across flats	Gage
102.	Inside diameter, max	Gage
103.	Outside diameter forward of flange, min	Gage
104.	Pitch diameter of thread, min	Gage
105.	Major diameter of thread, min	Gage
106.	Length from flange to shoulder, min	Gage

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107. Thickness of flange, min.....Gage
108. Material (see 3.1).....Visual
109. Objective evidence (see 6.2) available for protective finish in accordance with dwg 8862512.....Visual

Minor:
201. Radius present.....Visual
* 202. Stamping present and performed before plating.....Visual
203. Corner broken.....Visual
204. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.2 Retainer (see dwg 8862512 covering a detail of dwg 8863402).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major: None defined.

Minor:
201. Paint present no bare spot.....Visual
202. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.3 Retainer assembly (see dwg 8863402 covering a detail of dwg 8862511).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Objective evidence (see 6.2) available for sleeve bearing in accordance with MS35771.....Visual
102. Bearing tight in retainer.....Manual
103. Length from end of bearing to end of retainer.....*Measure

*Measure with suitable scale.

Minor:
201. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.4 Bearing, roller, thrust (see dwg 8862510 covering a detail of dwg 8862511).

Categories	Characteristics	Method of Inspection

Critical: None defined.

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Major:
101. Inside diameter, max.....Gage
102. Objective evidence (see 6.2) available for part in accordance with dwg 8862510.....Visual

Minor:
201. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.5 Plate, support (see dwg 8857430 covering a detail of dwg 8862511) (before painting).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Pitch diameter of thread, max.....Gage
102. Minor diameter of thread, max.....Gage
103. Small bottom diameter.....Gage
104. Small outside diameter, min.....Gage
105. Total thickness, min.....Gage
106. Width of arms, min.....Gage
107. Depth of undercut on arms, max.....Gage
108. Outside diameter, min.....Gage
109. Material (see 3.1).....Visual
110. Objective evidence (see 6.2) available for protective finish in accordance with dwg 8857430.....Visual

Minor:
201. Radius present.....Visual
* 202. Stamping present and performed before plating.....Visual
203. Corners broken.....Visual
204. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.6 Plate, support (see dwg 8857430 covering a detail of dwg 8862511)

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major: None defined.

Minor:
201. Paint present no bare spot.....Visual
202. Workmanship (see 3.7).....Visual/Manual

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4.3.2.1.7 Rod (see dwg 8857515 covering a detail of dwg 8857427) and (dwg 8839642 covering a detail of dwg 8839640).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Length of rod, min	Gage
102.	Pitch diameter of thread forward end, min	Gage
103.	Major diameter of thread forward end, min	Gage
104.	Length of thread forward end, min	Gage
105.	Length of flats forward end, max	Gage
106.	Thickness of rod through flats	Gage
107.	Diameter of rod, min	Gage
108.	Length of hexagonal section, min	Gage
109.	Width across flats of hexagonal section	Gage
110.	Material (see 3.1)	Visual
Minor:		
201.	Workmanship (see 3.7)	Visual

4.3.2.1.8 Plug (see dwg 8857514 covering a detail of dwg 8857427).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Concentricity of inside diameter	Gage
102.	Large inside diameter	Gage
103.	Concentricity of outside diameter	Gage
104.	Outside diameter	Gage
105.	Length of large inside diameter	Gage
106.	Depth of large thread cavity	Gage
107.	Width of large thread undercut, max	Gage
108.	Diameter of large thread undercut, max	Gage
109.	Pitch diameter of small inside thread, max	Gage
110.	Minor diameter of small inside thread, max	Gage
111.	Concentricity of small inside thread including Perpendicularity of small inside thread	Gage
112.	Total length, min	Gage
113.	Material (see 3.1)	Visual
Minor:		
201.	Workmanship (see 3.7)	Visual/Manual

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4.3.2.1.9 Plug and rod assembly (see dwg 8857427 covering a detail of dwg 8864331) (before painting).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Welding (see 4.3.3.1)	Visual
102.	Pitch diameter of rear thread, min (see dwg 8857515)	Gage
103.	Major diameter of rear thread, min (see dwg 8857515)	Gage
104.	Pitch diameter of thread in plug (see dwg 8857514)	Gage
105.	Minor diameter of thread in plug, max (see dwg 8857514)	Gage
106.	Objective evidence (see 6.2) available for protective finish in accordance with dwg 8857427	Visual
Minor:		
* 201.	Stamping present and performed before plating	Visual
202.	Workmanship (see 3.7)	Visual /Manual

4.3.2.1.10 Plug and rod assembly (see dwg 8857427 covering a detail of dwg 8864331) (after painting).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major: None defined.		
Minor:		
201.	Paint present no bare spot	Visual
202.	Workmanship (see 3.7)	Visual /Manual

4.3.2.1.11 Gusset (see dwg 8862527 covering a detail of dwg 8862525).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Material (see 3.1)	Visual
Minor:		
201.	Length of gusset, min	Gage

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202. Height of gusset, min.....Gage
203. Thickness of gusset, min.....Gage
204. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.12 Transducer, threaded (see dwg 8862526 covering a detail of dwg 8862525).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Pitch diameter of thread, max.....Gage
102. Minor diameter of thread, max.....Gage
103. Width across flats of hexagonal section.....Gage
104. Diameter of counterbore, max.....Gage
105. Depth of counterbore, max.....Gage
106. Length of flange, min.....Gage
107. Material (see 3.1).....Visual

Minor:

201. Workmanship (see 3.7).....Visual/Manual

* 4.3.2.1.13 Threaded transducer assembly (see dwg 8862525 sheet 1 covering a detail of dwg 8864331 or dwg 8857225) (before painting).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Welding (see 4.3.3.1).....Visual
102. Location of gussets.....Visual
103. Objective evidence (see 6.2) available for protective finish in accordance with dwg 8862525.....Visual

Minor:

* 201. Stamping present and performed before plating.....Visual
202. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.14 Threaded transducer assembly (see dwg 8862525 sheet 1 covering a detail of dwg 8864331 or dwg 8857225).

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Categories	Characteristics	Method of inspection

Critical: None defined.

Major: None defined.

Minor:

201. Paint present no bare spot.....Visual
202. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.15 Threaded transducer (alternate) (see dwg. 8862525 sheet 2 covering a detail of dwgs. 8857225 and 8864331) (before painting).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

101. Pitch diameter of thread, max.....Gage
102. Minor diameter of thread, max.....Gage
103. Length of rear inside diameter, max.....Gage
104. Rear inside diameter, max.....Gage
105. Length of flange, min.....Gage
106. Width across flats of hex.....Gage
107. Material (see 3.1).....Visual
108. Objective evidence (see 6.2) available for protective finish in accordance with dwg. 8862525.....Visual

Minor:

* 201. Stamping present and performed before plating.....Visual
202. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.16 Threaded transducer (alternate) (see dwg. 8862525 sheet 2 covering a detail of dwgs. 8857225 and 8864331.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major: None defined.

Minor:

201. Paint present no bare spot.....Visual
202. Workmanship (see 3.7).....Visual/Manual

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4.3.2.1.17 Rod, extension (see dwg. 8863343 covering a detail of dwgs. 8857225 and 8864331).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Width across flats of hexagonal end	Gage
102.	Length of flats of hexagonal end, min	Gage
103.	Pitch diameter of rear thread	Gage
104.	Major diameter of rear thread, min	Gage
105.	Length of large outside diameter including chamfer, min	Gage
106.	Outside diameter forward end, min	Gage
107.	Pitch diameter of forward thread	Gage
108.	Minor diameter of forward thread, max	Gage
109.	Depth of forward thread, min	Gage
110.	Depth of hole forward end, max	Gage
111.	Material (see 3.1)	Visual
112.	Objective evidence (see 6.2) available for protective finish in accordance with dwg. 8863343	Visual
Minor:		
201.	Chamfer or bevel present	Visual
* 202.	Stamping present and performed before plating	Visual
203.	Workmanship (see 3.7)	Visual/Manual

4.3.2.1.18 Wrench, bearing (see dwg. 8862529 covering a detail of dwg. 8864331 or dwg. 8857225) (before painting).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Thickness, min	Gage
102.	Total width wrench portion, min	Gage
103.	Total width inside wrench portion	Gage
104.	Outside radius of wrench portion, min	Gage
105.	Inside diameter of wrench portion, min	Gage
106.	Material (see 3.1)	Visual
107.	Objective evidence (see 6.2) available for protective finish in accordance with dwg. 8862529	Visual

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Minor:

201. Radii or chamfer present.....Visual
* 202. Stamping present and performed before plating.....Visual
203. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.19 Wrench, bearing (see dwg. 8862529 covering a detail of dwg. 8864331 or dwg. 8857225).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major: None defined.

Minor:

201. Paint present no bare spot.....Visual
202. Workmanship (see 3.7).....Visual/Manual

* 4.3.2.1.20 Wrench, torque (see dwg. 8839639 covering a detail of dwg. 8864331 or dwg. 8857225) (before painting).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

101. Thickness, min.....Gage
102. Location large hexagon opening.....Gage
103. Location small hexagon opening.....Gage
104. Width across flats large hexagon.....Gage
105. Width across flats small hexagon.....Gage
106. Material (see 3.1).....Visual
107. Objective evidence (see 6.2) available for protective finish in accordance with dwg. 8839639.....Visual

Minor:

* 201. Stamping present and performed before plating.....Visual
202. Corners broken.....Visual
203. Workmanship (see 3.7).....Visual/Manual

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4.3.2.1.21 Wrench, torque (see dwg. 8839639 covering a detail of dwg. 8864331 or dwg. 8857225).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major. None defined.		
Minor:		
201.	Paint present no bare spot	Visual
202.	Workmanship (see 3.7)	Visual/Manual

4.3.2.1.22 Plug (see dwg. 8839641 covering a detail of dwg. 8839640).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Length of pilot diameter of outside thread, max	Gage
102.	Length of undercut of outside thread, max	Gage
103.	Total length, min	Gage
104.	Location of 2 holes	Gage
105.	Pitch diameter inside thread, max	Gage
106.	Minor diameter inside thread, max	Gage
107.	Material (see 3.1)	Visual
* 108.	Depth of 2 small holes, max	Gage
* 109.	Diameter of 2 small holes, max	Gage
Minor:		
201.	Radius or chamfer present	Visual
202.	Workmanship (see 3.7)	Visual/Manual

4.3.2.1.23 Pin (see dwg. 8857302 covering a detail of dwg. 8839640).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Total length, min	Gage
102.	Material (see 3.1)	Visual
103.	Diameter, min	Gage

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Minor:

201. Chamfer present.....Visual
202. Workmanship (see 3.7).....Visual/Manual

* 4.3.2.1.24 Plug and rod assembly (see dwg. 8839640 covering a detail of dwg. 8857225) (before painting).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

101. Pin projection beyond plug.....Gage
102. Pitch diameter of rear thread, min (see dwg. 8839642)....Gage
103. Major diameter of rear thread, min (see dwg. 8839642)..Gage
104. Pitch diameter of forward thread (see dwg. 8839641)....Gage
105. Major diameter of forward thread (see dwg. 8839641)....Gage
106. Diameter of pins max (see dwg. 8857302).....Gage
* 107. Welding (see 4.3.3.3).....Visual
* 108. Weld below forward surface.....Visual
109. Objective evidence (see 6.2) available for protective finish in accordance with dwg. 8839640.....Visual

Minor:

* 201. Stamping present and performed before plating.....Visual
* 202. Workmanship (see 3.7).....Visual/Manual

* 4.3.2.1.25 Plug and rod assembly (see dwg. 8839640 covering a detail of dwg. 8857225) (after painting).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major: None defined.

Minor:

201. Paint present no bare spot.....Visual
202. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.26 Tool, shell removal assembly H-4172 (see dwg. 8857225) and tool, shell removal assembly XH-4196 (see dwg. 8864331) except assembly of roller bearing.

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Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Transducer moves freely on rod	Manual
102.	Protective finish present no bare spot	Visual
Minor:		
201.	No part missing, correct assembly except roller thrust bearing	Visual
202.	Workmanship (see 3.7)	Visual/Manual

4.3.2.1.27 Box, packaging, ammo for tool, shell removal assembly XH-4196 (see dwg. 8863248) and box, packaging, ammo, for tool, shell removal assembly: H-4172 (see dwg. 8863247) before closing cover.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	No component missing	Visual
102.	Roller bearing packed in grease in hermetically sealed barrier bag	Visual
Minor:		
201.	Workmanship (see 3.7)	Visual/Manual

4.3.2.1.28 Box packaging, ammo, for tool, shell removal assembly XH-4196 (see dwg. 8863248) and box packaging, ammo, for tool, shell removal assembly: H-4172 (see dwg. 8863247).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	No strapping missing, broken or loose	Visual/Manual
102.	No hardware, missing, broken or loose	Visual/Manual
103.	No board broken, exposing contents	Visual
104.	Marking	Visual
105.	Car seal, present and sealed	Visual

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Minor:

201. Finish present no bare spot.....Visual
202. Hardware properly fastened.....Visual
203. Steel strapping properly located.....Visual
204. Workmanship (see 3.7).....Visual

4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in the order specified.

	Test	Test Classification
a.	Tensile	Major
b.	Static loading capacity	Major
c.	Welding	Major
d.	Function test	Major
* e.	Paint adhesion	Minor

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Tensile (see dwgs. 8857514 and 8863343). A sample shall be selected at random from each lot of material and tested in accordance with 4.4.1. If any test specimen fails to comply with drawing requirements, the lot of material shall be rejected.

4.3.3.2 Static loading capacity (see 3.4). If a certificate of compliance certifying that all drawing requirements have been complied with has not been supplied by the bearing manufacturer, each bearing shall be tested in accordance with paragraph 4.4.2 and comply with 3.4. Any bearing which does not meet requirements will be removed from the lot.

* 4.3.3.3 Welding (see dwgs. 8857427 and 8839640). Welding shall be in accordance with MIL-STD-1261, Class II procedure. In addition each weld shall be inspected in accordance with MIL-I-6866 or MIL-I-6868. Any weld which shows evidence of cracks shall reject the item.

4.3.3.4 Functional test (see 3.5). Each packing box shall be tested in accordance with paragraph 4.4.3 and comply with 3.5. Any packing box which does not meet the requirements shall be removed from the lot.

* 4.3.3.5 Paint adhesion (see 3.6). Each part or assembly shall be tested as specified in 4.4.4. If any part or assembly fails to comply with 3.6, the part or assembly shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

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4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-8857225 or 8864331 as applicable. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Tensile (see 4.3.3.1). A minimum of 2 test specimens shall be prepared and tested in accordance with Federal Test Method Standard No. 151, Method 211.1 on approved test equipment.

4.4.2 Static load capacity of bearing (see 3.4). The bearing shall be subjected to a minimum load of 120,000 pounds on a Munitions Command approved loading device. The diameter of each roller shall be measured at points of contact.

4.4.3 Function test (see 3.5). Each packing box shall be tested by placing the components of the shell removal tool assembly into the box in their correct positions.

4.4.4 Paint adhesion (see 3.6). Paint adhesion shall be tested in accordance with Paint adhesion, section 4, TT-C-490.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The tool shall be packaged in accordance with dwgs. 8863247 or 8863248 as applicable using the specified moisture barrier.

5.1.2 Level C. The tool shall be packaged in accordance with dwgs. 8863247 or 8863248 as applicable except that the moisture barrier may be omitted.

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5.2 Packing.

5.2.1 Level A and C. The tool shall be packed in accordance with dwgs. 8863247 or 8863248 as applicable.

5.3 Marking. Marking shall be as specified on dwgs. 8797272 and 8863247 or 8863248 as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval or equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

* 6.4 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.5 Instructions for first article inspection (see 4.2.1). Parts may be used in subassemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government. Subassemblies may be used in final assemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government.

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